EASY · ADVANCED MATH · ANSWER KEY

Nonlinear functions

02

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

6

The correct answer is 6. An x-intercept of a graph is a point on the graph where it intersects the x-axis, or where the value of y is 0. The graph shown intersects the x-axis at the point $6, 0$. Therefore, the x-coordinate of the x-intercept of the graph shown is 6.

Q2

C

C. 36

Choice C is correct. It's given that $fx = 8\sqrt{x}$. Substituting 48 for fx in this equation yields $48 = 8\sqrt{x}$. Dividing both sides of this equation by 8 yields $6 = \sqrt{x}$. This can be rewritten as $\sqrt{x} = 6$. Squaring both sides of this equation yields $x = 36$. Therefore, the value of x for which $fx = 48$ is 36.

Choice A is incorrect. If $x = 6$, $fx = 8\sqrt{6}$, not 48.

Choice B is incorrect. If $x = 8$, $fx = 8\sqrt{8}$, not 48.

Choice D is incorrect. If $x = 64$, $fx = 8\sqrt{64}$, which is equivalent to 64, not 48.

Q3

C

C. 20,000

Choice C is correct. It's given that x represents years after 2010. Therefore, 2010 is represented by $x = 0$. On the model shown, the point with an x-coordinate of 0 has a y-coordinate of $20{,}000$. Thus, the model estimates that in 2010, the city had $20{,}000$ residents.

Choice A is incorrect. This is the value of x that represents the year 2010.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is approximately the number of residents the model estimates the city had in 2014, not 2010.

Q4

7

The correct answer is 7. It's given that the parabola intersects the y-axis at the point x, y. The graph shows that the parabola intersects the y-axis at the point $0, 7$. Therefore, the value of y is 7.

Q5

A

A. 118

Choice A is correct. Substituting 2 for x in the given equation yields $k(2) = 2^3 + 110$, which is equivalent to $k(2) = 8 + 110$, or $k(2) = 118$. Therefore, the value of $k(x)$ when $x = 2$ is 118.

Choice B is incorrect. This is the value of $k(x)$ when $x = 2$ if $k(x) = 3x + 110$, not $k(x) = x^3 + 110$.

Choice C is incorrect. This is the value of $k(x)$ when $x = 2$ if $k(x) = x + 3 + 110$, not $k(x) = x^3 + 110$.

Choice D is incorrect. This is the value of $k(x)$ when $x = 0$, not $x = 2$.

Q6

B

B. The estimated number of marine mammals in the area was 1,800 when the study began.

Choice B is correct. The function P gives the estimated number of marine mammals in a certain area, where t is the number of years since a study began. Since the value of $P0$ is the value of Pt when $t = 0$, it follows that $P0 = 1{,}800$ means that the value of Pt is 1,800 when $t = 0$. Since t is the number of years since the study began, it follows that $t = 0$ is 0 years since the study began, or when the study began. Therefore, the best interpretation of $P0 = 1{,}800$ in this context is the estimated number of marine mammals in the area was 1,800 when the study began.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7

C

C. 5

Choice C is correct. It’s given that the function g is defined by $gx = \sqrt{8x+1}$. Substituting 3 for x in the given function yields $g3 = \sqrt{83+1}$, which is equivalent to $g3 = \sqrt{25}$, or $g3 = 5$. Therefore, the value of $g3$ is 5.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $83 + 1$, not $\sqrt{83+1}$.

Q8

A

A. $\frac{16}{17}$

Choice A is correct. It's given that $fx = \frac{16}{x}$. Substituting 17 for x in this function yields $f17 = \frac{16}{17}$. Therefore, when $x = 17$, the value of fx is $\frac{16}{17}$.

Choice B is incorrect. This is the value of the reciprocal of fx when $x = 17$.

Choice C is incorrect. This is the value of fx when $x = 1$.

Choice D is incorrect. This is the value of x when $x = 17$.

Q9 C

C. 309

Choice C is correct. It's given that function g is defined by $g\left(x\right) = \sqrt{x} + 300$. Substituting 81 for x in function g yields $g\left(81\right) = \sqrt{81} + 300$, which is equivalent to $g\left(81\right) = 9 + 300$, or $g\left(81\right) = 309$. Therefore, the value of $g\left(x\right)$ when $x = 81$ is 309.

Choice A is incorrect. This is the value of $\sqrt{81}$, not $\sqrt{81} + 300$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $81 + 300$, not $\sqrt{81} + 300$.

Q10 A

A. 4

Choice A is correct. It's given that $gx = x^2 + 9$. Substituting 25 for gx in this equation yields $25 = x^2 + 9$. Subtracting 9 from both sides of this equation yields $16 = x^2$. Taking the square root of each side of this equation yields $x = \pm 4$. It follows that $gx = 25$ when the value of x is 4 or -4. Only 4 is listed among the choices.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank